

Ch24: 4 , 9 , 10 , 24 , 34 , 48 , 53 , 67

Basic score 4` + problem 9 2`+ problem 24 2`+ problem 34 2`

4. (a) Consider an infinitesimal segment of the rod, located between x and $x + dx$. It has length dx and contains charge $dq = \lambda dx$, where $\lambda = Q/L$ is the linear charge density of the rod. Its distance from P_1 is $d + x$ and the potential it creates at P_1 is

$$\begin{aligned} dV &= \frac{1}{4\pi\epsilon_0} \frac{dq}{d+x} = \frac{1}{4\pi\epsilon_0} \frac{\lambda dx}{d+x} \Rightarrow V = \frac{1}{4\pi\epsilon_0} \int_0^L \frac{\lambda dx}{d+x} \\ \Rightarrow V &= \frac{Q}{4\pi\epsilon_0 L} \ln\left(\frac{L+d}{d}\right) = \frac{(8.99 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2)(43.6 \times 10^{-15} \text{ C})}{0.135 \text{ m}} \ln\left(1 + \frac{0.135 \text{ m}}{d}\right) \\ &= (2.90 \times 10^{-3} \text{ V}) \ln\left(1 + \frac{0.135 \text{ m}}{d}\right) \end{aligned}$$

(b) We differentiate the potential with respect to x to find the x component of the electric field:

$$\begin{aligned} V &= \frac{Q}{4\pi\epsilon_0 L} \ln\left(\frac{L-x}{-x}\right) = \frac{Q}{4\pi\epsilon_0 L} \ln\left(\frac{x-L}{x}\right) \\ E_x &= -\frac{\partial V}{\partial x} = -\frac{Q}{4\pi\epsilon_0 L} \frac{\partial}{\partial x} \ln\left(\frac{x-L}{x}\right) = -\frac{Q}{4\pi\epsilon_0 L} \frac{\partial}{\partial x} \ln\left(1 - \frac{L}{x}\right) = -\frac{Q}{4\pi\epsilon_0 L} \frac{x}{x-L} \left(\frac{L}{x^2}\right) \\ &= -\frac{Q}{4\pi\epsilon_0 x(x-L)} = -\frac{(8.99 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2)(43.6 \times 10^{-15} \text{ C})}{x(x-0.135 \text{ m})} = -\frac{(3.92 \times 10^{-4} \text{ N} \cdot \text{m}^2/\text{C})}{x(x-0.135 \text{ m})} \end{aligned}$$

(c) Since $E_x < 0$, its direction relative to the positive x axis is 180° .

(d) At $x = -d = -6.60 \text{ cm}$, we obtain

$$E_x = -\frac{(3.92 \times 10^{-4} \text{ N} \cdot \text{m}^2/\text{C})}{(-0.0660 \text{ m})(-0.0660 \text{ m} - 0.135 \text{ m})} = -0.0295 \text{ N/C}.$$

(e) Consider two points an equal infinitesimal distance on either side of P_1 , along a line that is perpendicular to the x axis. The difference in the electric potential divided by their separation gives the transverse component of the electric field. Since the two points are situated symmetrically with respect to the rod, their potentials are the same and the potential difference is zero. Thus, the transverse component of the electric field E_y is zero.

9. (a) All the charge is the same distance R from C , so the electric potential at C is

$$V = \frac{1}{4\pi\epsilon_0} \left(\frac{Q_1}{R} - \frac{6Q_1}{R} \right) = -\frac{5Q_1}{4\pi\epsilon_0 R} \quad 0.5 \text{ point}$$

$$= -\frac{5(8.99 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2)(7.07 \times 10^{-12} \text{ C})}{8.20 \times 10^{-2} \text{ m}} = -3.88 \text{ V} \quad 0.5 \text{ point}$$

1 point in total

where the zero was taken to be at infinity.

(b) All the charge is the same distance from P . That distance is $\sqrt{R^2 + D^2}$, so the electric potential at P is

$$V = \frac{1}{4\pi\epsilon_0} \left[\frac{Q_1}{\sqrt{R^2 + D^2}} - \frac{6Q_1}{\sqrt{R^2 + D^2}} \right] = -\frac{5Q_1}{4\pi\epsilon_0 \sqrt{R^2 + D^2}} \quad 0.5 \text{ point}$$

$$= -\frac{5(8.99 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2)(7.07 \times 10^{-12} \text{ C})}{\sqrt{(8.20 \times 10^{-2} \text{ m})^2 + (2.05 \times 10^{-2} \text{ m})^2}} \quad 1 \text{ point in total}$$

$$= -3.76 \text{ V.} \quad 0.5 \text{ point}$$

10. The change in electric potential energy of the electron-shell system as the electron starts from its initial position and just reaches the shell is $\Delta U = (-e)(-V) - 0 = eV$.

Thus from $\Delta U = K = \frac{1}{2} m_e v_i^2$ we find the initial electron speed to be

$$v_i = \sqrt{\frac{2\Delta U}{m_e}} = \sqrt{\frac{2eV}{m_e}} = \sqrt{\frac{2(1.6 \times 10^{-19} \text{ C})(170 \text{ V})}{9.11 \times 10^{-31} \text{ kg}}} = 7.73 \times 10^6 \text{ m/s.}$$

$$\mathbf{24.} \quad W_a = \Delta U = q\Delta V = q(V_f - V_i) = e \left(\frac{p \cos \theta}{4\pi\epsilon_0 r^2} - \frac{p \cos 0^\circ}{4\pi\epsilon_0 r^2} \right) = \frac{ep}{4\pi\epsilon_0 r^2} (\cos \theta - 1) \quad 1 \text{ point}$$

with $r = 20 \times 10^{-9} \text{ m}$. For $\theta = 180^\circ$ the graph indicates $W_a = -2.0 \times 10^{-30} \text{ J}$, (0.5 point) from which we can determine p . The magnitude of the dipole moment is therefore

$$p = \frac{4\pi\epsilon_0 r^2 W_a}{e(\cos \theta - 1)} = \frac{4\pi(8.85 \times 10^{-12})(20 \times 10^{-9})^2 (-2.0 \times 10^{-30})}{1.6 \times 10^{-19} (\cos 180^\circ - 1)} = 2.8 \times 10^{-37} \text{ C} \cdot \text{m} \quad 0.5 \text{ point}$$

34. In the “inside” region between the plates, the individual fields (given by Eq. 24-13) are in the same direction ($-\hat{i}$):

$$\vec{E}_{\text{in}} = -\left(\frac{50 \times 10^{-9} \text{ C/m}^2}{2(8.85 \times 10^{-12} \text{ C}^2/\text{N} \cdot \text{m}^2)} + \frac{25 \times 10^{-9} \text{ C/m}^2}{2(8.85 \times 10^{-12} \text{ C}^2/\text{N} \cdot \text{m}^2)}\right) \hat{i} = -(4.2 \times 10^3 \text{ N/C}) \hat{i}.$$

0.5 point

In the “outside” region where $x > 0.5 \text{ m}$, the individual fields point in opposite directions:

$$\vec{E}_{\text{out}} = -\frac{50 \times 10^{-9} \text{ C/m}^2}{2(8.85 \times 10^{-12} \text{ C}^2/\text{N} \cdot \text{m}^2)} \hat{i} + \frac{25 \times 10^{-9} \text{ C/m}^2}{2(8.85 \times 10^{-12} \text{ C}^2/\text{N} \cdot \text{m}^2)} \hat{i} = -(1.4 \times 10^3 \text{ N/C}) \hat{i}.$$

0.5 point

Therefore, we have

$$\begin{aligned} \Delta V = V_1 - V_0 &= -\int_0^1 \vec{E} \cdot d\vec{s} = -\left\{ \int_0^{0.5} \vec{E}_{\text{in}} \cdot dx \hat{i} - \int_{0.5}^1 \vec{E}_{\text{out}} \cdot dx \hat{i} \right\} = -\left\{ \int_0^{0.5} -|\vec{E}_{\text{in}}| \cdot dx - \int_{0.5}^1 -|\vec{E}_{\text{out}}| \cdot dx \right\} \\ &= (4.2 \times 10^3)(0.5) + (1.4 \times 10^3)(0.5) = 2.8 \times 10^3 \text{ V}. \end{aligned}$$

1 point (过程结果各 0.5 分，结果为负不扣分)

Alternatively, we also can calculate the potential difference created by each charged plate and then sum them up.

For the plate with surface charge density: $\sigma_1 = 50 \text{ nC/m}^2$:

$$\vec{E}_1 = -\frac{\sigma_1}{2\epsilon_0} \hat{i} = -\left(\frac{50 \times 10^{-9} \text{ C/m}^2}{2(8.85 \times 10^{-12} \text{ C}^2/\text{N} \cdot \text{m}^2)}\right) \hat{i}$$

$$\Delta V_1 = \frac{\sigma_1}{2\epsilon_0} d = \frac{50 \times 10^{-9} \text{ C/m}^2 \times 1 \text{ m}}{2(8.85 \times 10^{-12} \text{ C}^2/\text{N} \cdot \text{m}^2)} = 2.82 \times 10^3 \text{ V}$$

1 point

For the plate with surface charge density: $\sigma_2 = -25 \text{ nC/m}^2$:

$$\vec{E}_2 = -\frac{\sigma_2}{2\epsilon_0} \hat{i} \text{ for } x < 0.5 \text{ m and } \frac{\sigma_2}{2\epsilon_0} \hat{i} \text{ for } x > 0.5 \text{ m}$$

$$\text{So } \Delta V_2 = \frac{\sigma_2}{2\epsilon_0} \times 0.5 - \frac{\sigma_2}{2\epsilon_0} \times 0.5 = 0$$

0.5 point

So the potential difference created by the two charge plates is:

$$\Delta V = \Delta V_1 + \Delta V_2 = 2.82 \times 10^3 \text{ V}$$

0.5 point

48. Since the charge distribution is spherically symmetric we may write

$$E(r) = \frac{1}{4\pi\epsilon_0} \frac{q_{\text{enc}}}{r^2},$$

where q_{enc} is the charge enclosed in a sphere of radius r centered at the origin.

(a) For $r = 4.00 \text{ m}$, $R_2 = 1.00 \text{ m}$, and $R_1 = 0.500 \text{ m}$, with $r > R_2 > R_1$ we have

$$E(r) = \frac{q_1 + q_2}{4\pi\epsilon_0 r^2} = \frac{(8.99 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2)(3.00 \times 10^{-6} \text{ C} + 1.00 \times 10^{-6} \text{ C})}{(4.00 \text{ m})^2} = 2.25 \times 10^3 \text{ V/m}.$$

(b) For $R_2 > r = 0.700 \text{ m} > R_1$,

$$E(r) = \frac{q_1}{4\pi\epsilon_0 r^2} = \frac{(8.99 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2)(3.00 \times 10^{-6} \text{ C})}{(0.700 \text{ m})^2} = 5.50 \times 10^4 \text{ V/m}.$$

(c) For $R_2 > R_1 > r$, the enclosed charge is zero. Thus, $E = 0$.

The electric potential may be obtained using :

$$V(r) - V(r') = -\int_r^{r'} E(r) dr.$$

(d) For $r = 4.00 \text{ m} > R_2 > R_1$, we have

$$V(r) = \frac{q_1 + q_2}{4\pi\epsilon_0 r} = \frac{(8.99 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2)(3.00 \times 10^{-6} \text{ C} + 1.00 \times 10^{-6} \text{ C})}{(4.00 \text{ m})} = 8.99 \times 10^3 \text{ V}.$$

(e) For $r = 1.00 \text{ m} = R_2 > R_1$, we have

$$V(r) = \frac{q_1 + q_2}{4\pi\epsilon_0 r} = \frac{(8.99 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2)(3.00 \times 10^{-6} \text{ C} + 1.00 \times 10^{-6} \text{ C})}{(1.00 \text{ m})} = 3.60 \times 10^4 \text{ V}.$$

(f) For $R_2 > r = 0.700 \text{ m} > R_1$,

$$\begin{aligned} V(r) &= \frac{1}{4\pi\epsilon_0} \left(\frac{q_1}{r} + \frac{q_2}{R_2} \right) = (8.99 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2) \left(\frac{3.00 \times 10^{-6} \text{ C}}{0.700 \text{ m}} + \frac{1.00 \times 10^{-6} \text{ C}}{1.00 \text{ m}} \right) \\ &= 4.75 \times 10^4 \text{ V}. \end{aligned}$$

(g) For $R_2 > r = 0.500 \text{ m} = R_1$,

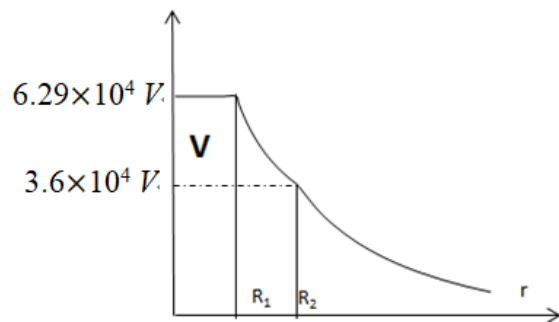
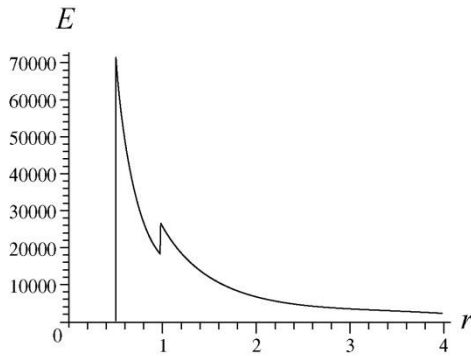
$$\begin{aligned} V(r) &= \frac{1}{4\pi\epsilon_0} \left(\frac{q_1}{r} + \frac{q_2}{R_2} \right) = (8.99 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2) \left(\frac{3.00 \times 10^{-6} \text{ C}}{0.500 \text{ m}} + \frac{1.00 \times 10^{-6} \text{ C}}{1.00 \text{ m}} \right) \\ &= 6.29 \times 10^4 \text{ V}. \end{aligned}$$

(h) For $R_2 > R_1 > r$,

$$\begin{aligned} V &= \frac{1}{4\pi\epsilon_0} \left(\frac{q_1}{R_1} + \frac{q_2}{R_2} \right) = (8.99 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2) \left(\frac{3.00 \times 10^{-6} \text{ C}}{0.500 \text{ m}} + \frac{1.00 \times 10^{-6} \text{ C}}{1.00 \text{ m}} \right) \\ &= 6.29 \times 10^4 \text{ V}. \end{aligned}$$

(i) At $r = 0$, the potential remains constant, $V = 6.29 \times 10^4 \text{ V}$.

(j) The electric field and the potential as a function of r are depicted below:



53. (a) The electric potential V at the surface of the drop, the charge q on the drop, and the radius R of the drop are related by $V = q/4\pi\epsilon_0 R$. Thus

$$R = \frac{q}{4\pi\epsilon_0 V} = \frac{(8.99 \times 10^9 \text{ N} \cdot \text{m}^2 / \text{C}^2)(30 \times 10^{-12} \text{ C})}{500 \text{ V}} = 5.4 \times 10^{-4} \text{ m}.$$

(b) After the drops combine the total volume is twice the volume of an original drop, so the radius R' of the combined drop is given by $(R')^3 = 2R^3$ and $R' = 2^{1/3}R$. The charge is twice the charge of original drop: $q' = 2q$. Thus,

$$V' = \frac{1}{4\pi\epsilon_0} \frac{q'}{R'} = \frac{1}{4\pi\epsilon_0} \frac{2q}{2^{1/3}R} = 2^{2/3}V = 2^{2/3}(500 \text{ V}) = 794 \text{ V}.$$

$$\text{c) } \sigma = \frac{q}{4\pi R^2} \Rightarrow \frac{\sigma_f}{\sigma_i} = \frac{2q}{q} \cdot \frac{R^2}{(2^{1/3}R)^2} = 1.26$$

67. The disk is uniformly charged. This means that when the full disk is present each quadrant contributes equally to the electric potential at P , so the potential at P due to a single quadrant is one-fourth the potential due to the entire disk. First find an expression for the potential at P due to the entire disk. We consider a ring of charge with radius r and (infinitesimal) width dr . Its area is $2\pi r dr$ and it contains charge $dq = 2\pi\sigma r dr$. All the charge in it is a distance $\sqrt{r^2 + D^2}$ from P , so the potential it produces at P is

$$dV = \frac{1}{4\pi\epsilon_0} \frac{2\pi\sigma r dr}{\sqrt{r^2 + D^2}} = \frac{\sigma r dr}{2\epsilon_0 \sqrt{r^2 + D^2}}.$$

The total potential at P is

$$V = \frac{\sigma}{2\epsilon_0} \int_0^R \frac{r dr}{\sqrt{r^2 + D^2}} = \frac{\sigma}{2\epsilon_0} \sqrt{r^2 + D^2} \Big|_0^R = \frac{\sigma}{2\epsilon_0} \left[\sqrt{R^2 + D^2} - D \right].$$

The potential V_{sq} at P due to a single quadrant is

$$V_{sq} = \frac{V}{4} = \frac{\sigma}{8\epsilon_0} \left[\sqrt{R^2 + D^2} - D \right] = \frac{(7.73 \times 10^{-15} \text{ C/m}^2)}{8(8.85 \times 10^{-12} \text{ C}^2/\text{N} \cdot \text{m}^2)} \left[\sqrt{(0.640 \text{ m})^2 + (0.450 \text{ m})^2} - 0.450 \text{ m} \right]$$
$$= 3.63 \times 10^{-5} \text{ V}.$$